

# Ground floor study

Where the garage stair lands, and the floor planned around it: a tall living room behind a pentagon of glass. Metres, to scale, house frame: x along the house (the valley at the bottom), z across (north to the right).

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**8.00 × 7.00**                      **x 5.70**  
over all; 7.40 × 6.40 clear, 47.4 m<sup>2</sup>      where the garage flight lands: the back wall line, 2.40 to the hill wall

**3.48 × 1.20**  
opening in both decks, from the 5th riser; headroom 2.00

**6.40 × 9.10**  
the panorama: one pentagon of glass, 5.90 at the walls, 9.10 at the apex

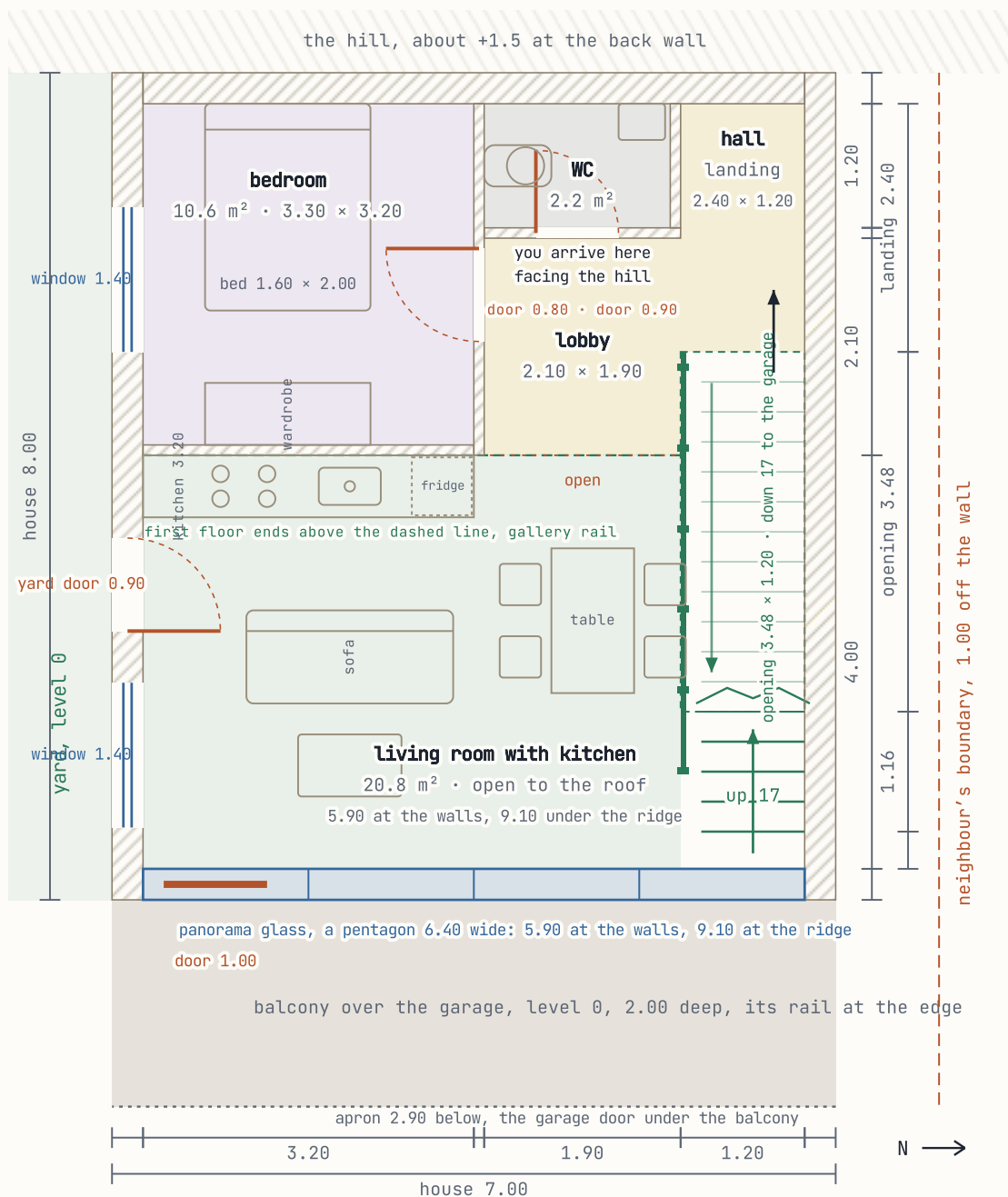
**45°**  
the roof: eaves at 5.60 on the walls, ridge at 9.10; the living room is open to it

**20.8 · 10.6 · 6.9 · 2.2**                      **21.8**  
living room with kitchen, bedroom, hall, toilet, m<sup>2</sup>      first floor left behind the tall room, m<sup>2</sup> clear: 3.40 × 6.40

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## Plan, 1 m above the floor

The hill is at the top, the valley and the balcony at the bottom, the yard to the left (south), the neighbour's boundary 1 m off the right (north) wall. Walls 0.30, partitions 0.10.



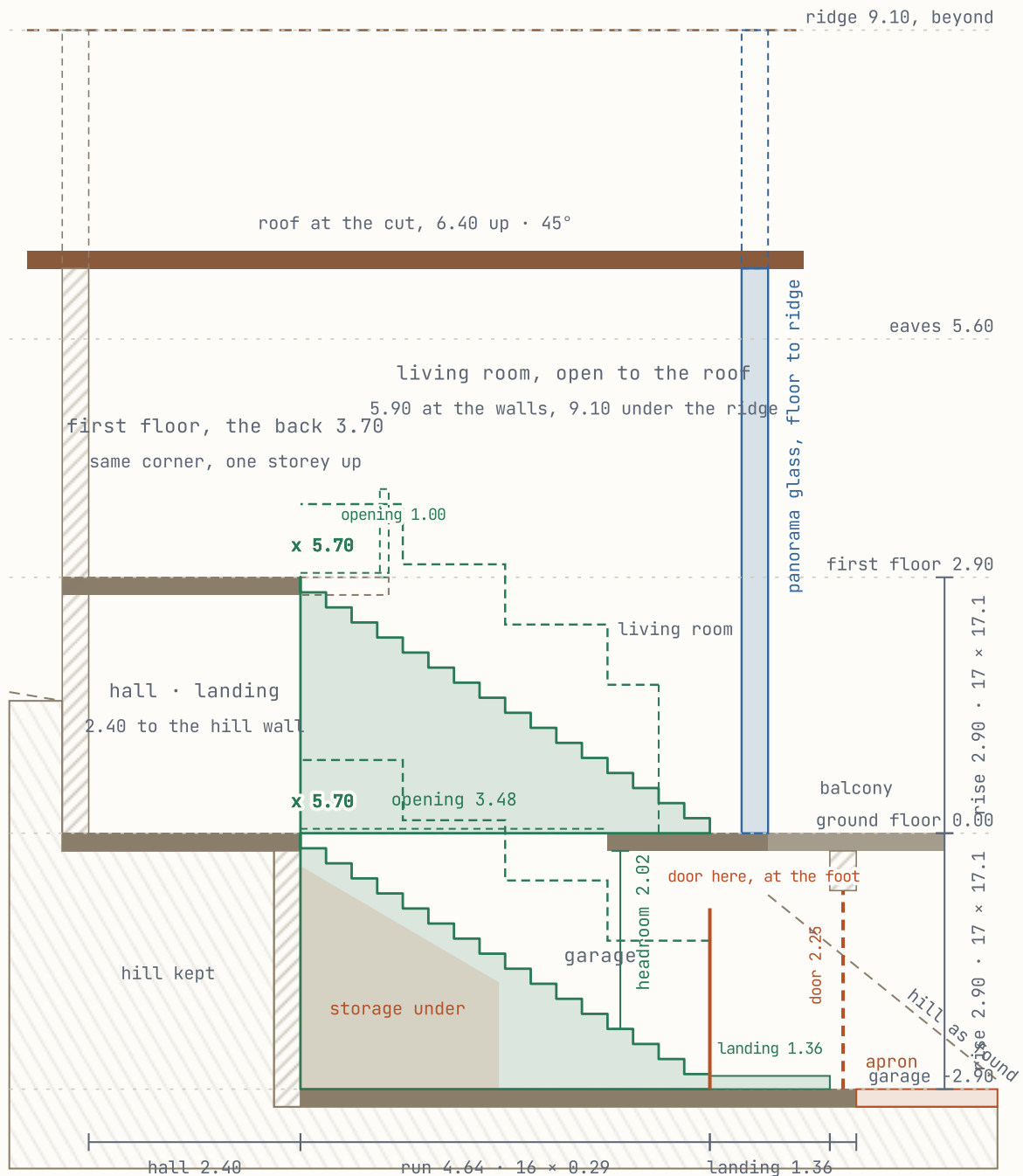
hall toilet bedroom living room with kitchen, open to the roof glass

- **The hall is where the stair lands.** You come up facing the hill at x 5.7, with 2.40 to the back wall; the lobby south of the landing has the toilet door on its west side and the bedroom door on its south side, and opens east into the living room. Nothing else is at the back, because the back has no light: the hill behind, the boundary beside.
- **The living room is the whole front,** 4.00 × 5.20, and it is tall: the first floor stops at x 6.7, on the bedroom wall, so from the kitchen to the glass the room rises past the first floor to the roof — 5.90 at the walls, 9.10 under the ridge. Its east wall is the panorama: one pentagon of glass, 6.40 wide, from the floor to the roof, with a 1.00 door to the balcony at its south end.

- **The entrance from the yard** is a 0.90 glazed door in the south wall beside the kitchen; a 1.40 window further along lights the sofa.
- **The kitchen** stands against the bedroom wall, 3.20 of counter with the fridge at the hall end. The table sits by the stair opening, the sofa faces the glass.
- **The bedroom** is  $3.30 \times 3.20$  in the south-west corner with a 1.40 window to the yard: a  $1.60 \times 2.00$  bed with its head to the hill wall, 0.70 past its foot to a 0.60 wardrobe. Its west wall and the toilet's are against the hill.
- **The stair up** stacks on the garage flight: same 17 risers, foot at  $x 10.34$  in the north-east corner, top at  $x 5.7$  again. It climbs inside the tall room along its north wall and arrives on the first floor's landing at the back. It is entered from the side, so treads 1 and 2 stay open to the room and the balustrade starts at riser 3.
- **The first floor keeps the back 3.70 m** ( $x 3-6.7$ ):  $3.40 \times 6.40$  clear,  $21.8 \text{ m}^2$ , a landing, a bedroom and a bathroom, with a gallery rail on its edge over the living room. If that is too little, a gallery out to  $x 7.9$  with the kitchen under it wins  $8 \text{ m}^2$  back.

## Where the garage stair lands: section along the flights, looking south

Cut 0.8 m from the north wall, through both flights. Garage floor -2.90, ground floor 0.00, first floor 2.90, eaves 5.60, ridge 9.10. The roof is cut where it is 6.40 up; the ridge is beyond.

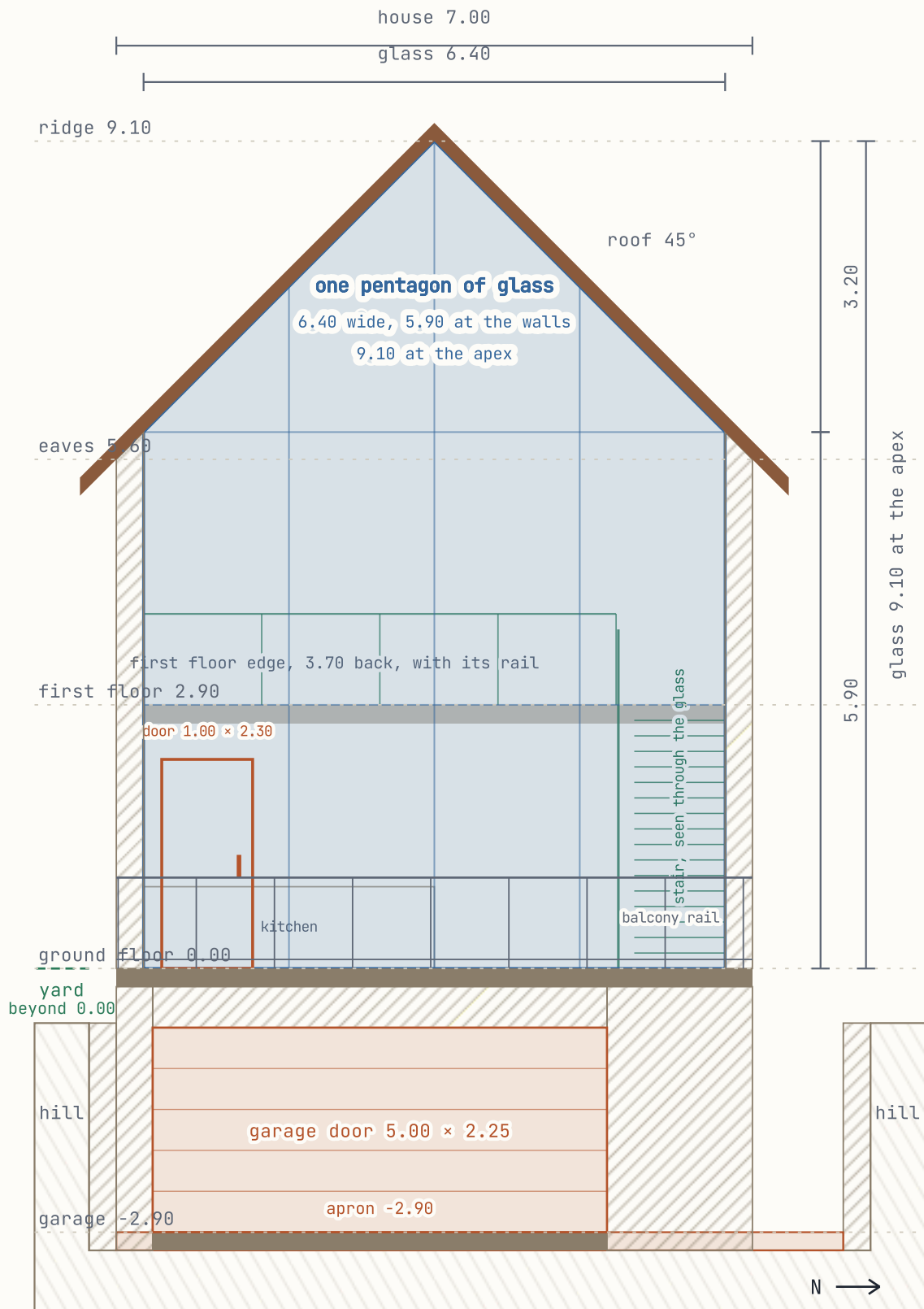


- **The top riser is the garage's back wall line, x 5.7:** the 17th step up lands on the ground floor over the garage's west wall, facing west into the hill side of the house with 2.40 m of floor ahead before the back wall. That is the landing, 2.40 × 1.20, in the floor's north-west corner.
- **The opening** runs from the 5th riser to the top, 3.48 × 1.20 along the north wall. The ground floor's deck stays over treads 1 to 4: 1.52 × 1.20 in the north-east corner, which is exactly the foot the next flight needs. So the stair to the first floor stacks on this one and arrives at the same spot one storey up; the first floor's deck ends at x 6.7, so its opening is the last metre, x 5.7–6.7, and the rest of the flight climbs in the open.

- **Headroom is 2.00** over the 4th tread under each deck, the minimum. Starting the opening one riser earlier would give 0.17 more and leave a 1.23 foot.
- **The door between garage and house** belongs at the foot of the garage flight, on the 1.36 landing inside the garage door, swinging onto that landing; a door at the top would open onto the stair.
- **What this does to the plan:** the whole north strip from x 5.7 to the east wall is stairs, so the hall, the toilet and the bedroom door gather at the back, and the front belongs to the living room — which, with the first floor ending at x 6.7, is open to the roof.

## East face, from the valley: the pentagon

Looking west at the glass, north to the right as in the plan. The garage door under the balcony, the balcony at the living room's floor, the glass from there to the ridge.



- **One pentagon of glass**, 6.40 wide between the north and south walls, 5.90 at the walls, 9.10 at the apex: about 47 m<sup>2</sup> of glazing, split by mullions every 1.60 and transoms at the first floor's level and the eaves line.

- **Seen through it:** the stair climbing the north wall, the first floor's edge 3.70 back with its rail, and the roof's underside.
- **The balcony door** is at the south end of the glass, 1.00 × 2.30, where the balcony and the yard meet at the house corner.

## Rooms

| ROOM                          | CLEAR                     | M <sup>2</sup> | LIGHT AND DOORS                                                                                                                                 |
|-------------------------------|---------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Living room with kitchen      | 4.00 × 5.20               | 20.8           | open to the roof, 5.90 to 9.10; the glass pentagon east with a 1.00 door to the balcony; yard door 0.90 and window 1.40 south; open to the hall |
| Bedroom                       | 3.30 × 3.20               | 10.6           | window 1.40 to the yard; door 0.90 from the lobby                                                                                               |
| Hall: landing and lobby       | 2.40 × 1.20 + 2.10 × 1.90 | 6.9            | the garage flight lands in it; borrows light from the living room                                                                               |
| Toilet                        | 1.20 × 1.80               | 2.2            | door 0.80 from the lobby; no window, mechanical vent                                                                                            |
| Stair opening                 | 3.48 × 1.20               | 4.2            | the garage flight below, the flight up above                                                                                                    |
| Stair foot, on the deck       | 1.52 × 1.20               | 1.8            | treads 1 to 4 of the flight up                                                                                                                  |
| Partitions                    |                           | 1.0            |                                                                                                                                                 |
| <b>Inside the outer walls</b> | <b>7.40 × 6.40</b>        | <b>47.4</b>    |                                                                                                                                                 |

## Still open

- The first floor: only the back 3.70 m is left under a tall living room — a landing, one bedroom and a bathroom (21.8 m<sup>2</sup> clear). A gallery out to x 7.9, with the kitchen in the low part under it, would win 8 m<sup>2</sup> back; the attic over it is another room if the roof is kept at 45°.
- The roof pitch: 45° drawn, which puts the apex at 9.10. At 35° it is 8.05 and the glass loses a metre.
- 47 m<sup>2</sup> of east glass: heat in winter and sun in summer want a plan — triple glazing, and shading or a deep roof overhang at the gable.
- The yard door: a plain door in the south wall by the kitchen, as drawn, or in the glass beside the balcony door.
- The door at the foot of the garage flight: self-closing, fire-rated where the rules ask.
- A north window: the wall is 1 m from the boundary; the rules decide.

- The west wall stands against the hill (about +1.5 there), so the bedroom's and the toilet's back walls are retaining walls and need tanking.
- Whether the outer walls sit inside the  $8 \times 7$  (drawn) or outside it: outside would give  $8.00 \times 7.00$  clear and push the east glass onto the balcony's first 0.30.